INCH-POUND

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MILITARY SPECIFICATION

TURNING ROLLS, WELDING, POWERED, IDLER, AND TILTING TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers turning rolls for supporting and rotating cylindrical workpieces in welding and related processes.

1.2 Classification. The turning rolls shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Powered

Size 1	-	3,000 lb. turning x	1,000 lb. load capacity.
Size 2	-	6,000 lb. turning x	2,000 lb. load capacity.
Size 2a	-	7,500 lb. turning x	2,500 lb. load capacity.
Size 3	-	10,000 lb. turning x	3,300 lb. load capacity.
Size 4	-	15,000 lb. turning x	5,000 lb. load capacity.
Size 5	-	20,000 lb. turning x	7,000 lb. load capacity.
Size 6	-	30,000 lb. turning x	10,000 lb. load capacity.
Size 7	-	40,000 lb. turning x	13,000 lb. load capacity.
Size 7a	-	45,000 lb. turning x	15,000 lb. load capacity.
Size 8	-	50,000 lb. turning x	17,000 lb. load capacity.
Size 9	-	60,000 lb. turning x	20,000 lb. load capacity.
Size 10	-	75,000 lb. turning x	25,000 lb. load capacity.
Size 10a	-	90,000 lb. turning x	30,000 lb. load capacity.
Size 11	-	100,000 lb. turning x	34,000 lb. load capacity.
Size 12	-	120,000 lb. turning x	40,000 lb. load capacity.
Size 13	-	150,000 lb. turning x	50,000 lb. load capacity.
Size 14	-	180,000 lb. turning x	60,000 lb. load capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - Idler

Size 1 - 1,000 lb. load capacity.
Size 2 - 2,000 lb. load capacity.
Size 2a- 2,500 lb. load capacity.
Size 3 - 3,300 lb. load capacity.
Size 4 - 5,000 lb. load capacity.
Size 5 - 7,000 lb. load capacity.
Size 6 - 10,000 lb. load capacity.
Size 7 - 13,000 lb. load capacity.
Size 7a- 15,000 lb. load capacity.
Size 8 - 17,000 lb. load capacity.
Size 9 - 20,000 lb. load capacity.
Size 10 - 25,000 lb. load capacity.
Size 10a- 30,000 lb. load capacity.
Size 11 - 34,000 lb. load capacity.
Size 12 - 40,000 lb. load capacity.
Size 13 - 50,000 lb. load capacity.
Size 14 - 60,000 lb. load capacity.

Type III - Tilting

Size 1 - 2,000 lb. load capacity - turning/tilting.
Size 2 - 3,000 lb. load capacity - turning/tilting.
Size 3 - 5,000 lb. load capacity - turning/tilting.
Size 4 - 10,000 lb. load capacity - turning/tilting.
Size 5 - 20,000 lb. load capacity - turning/tilting.
Size 6 - 30,000 lb. load capacity - turning/tilting.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

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MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

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(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete turning roll shall be subjected to first article inspection (see 6.4) in accordance with 4.4 .

3.2 Design. The turning roll shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The type 1 turning rolls shall be powered and have a turning capacity of not less than three times the support capacity. The turning rolls shall be so designed that one power driven turning roll and one or more mating idler turning rolls shall be capable of supporting and rotating a cylindrical workpiece for the size and capacity specified herein. The turning rolls and mating idler rolls shall be designed to comprise a turning roll set. The type II turning rolls shall be of the idler type. The type III tilting and turning roll shall be designed with a tilting base and means to allow the workpiece to rotate 360 degrees and the turning rolls shall be capable of tilting action of not less than 45 degrees for end welding. The turning rolls shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified, but are a part of the manufacturer's current design shall be included in the turning rolls furnished. When specified (see 6.2), the turning rolls shall be designed with the capability of rotating workpieces with unusual configurations under unbalanced load conditions. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the turning rolls shall be graduated in the same system.

3.2.2 Reclaimed materials. The turning rolls may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the turning rolls. Reclaimed materials shall have been collected or recovered from solid waste

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and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The turning rolls and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and electrical operating controls shall be located convenient to the operator's work stations.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the turning roll whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers, guards, and other safety devices. Covers,guards, and other safety devices normally furnished as standard on the manufacturer's commercial turning rolls supplied to the commercial market shall be provided for the point of operation and all other parts of the turning rolls that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the turning rolls (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the turning rolls. All guards shall allow access to the guarded parts.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the turning rolls when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the turning rolls, and its components, the turning roll shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions in effect on the date of the contract. The operation of the turning rolls at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Mercury restriction. The turning rolls shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the turning rolls.

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3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or turning rolls.

3.3 Construction. The turning rolls shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the turning rolls to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the turning rolls, related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The turning rolls shall consist of not less than a chassis assembly capable of supporting two or more turning roll wheel assemblies. The powered turning roll assemblies shall also be provided with all the necessary controls and hardware to comprise a complete turning roll set.

3.4.1 Chassis assembly. The chassis assembly shall be a steel weldment capable of maintaining original alignment of the rolls to minimize distortion. The base shall provide means for securing the chassis to the floor. Size 3 and larger turning rolls shall be provided with roll adjustment to accommodate workpieces of different diameters within the range specified in table I. Guards shall provide protection for all turning roll shafts in all adjusted positions. All rolls on a common chassis shall be aligned to such a degree as to assure maximum contact between the roll surfaces and the workpiece.

TABLE I. Turning roll capacities.

Turning roll type I and II size	Turning capacity (pounds) type I	Support capacity (pounds) type I and II	Motor horse-power type I only minimum	Roll wheel surface speed inches per minute (IPM)		Workpiece diameter	
				minimum	maximum	minimum (inches)	maximum (feet)
1	3,000	1,000	1/8	2	55	3	3
2	6,000	2,000	1/2	2	55	3	6
2a	7,500	2,500	1/2	2	50	4	12
3	10,000	3,333	1/2	2	50	4	6
4	15,000	5,000	1/2	2	50	6	13
5	20,000	6,666	3/4	2	50	6	13
6	30,000	10,000	1-1/2	2	40	8	18
7	40,000	13,333	1-1/2	1.5	40	8	18
7a	45,000	15,000	1-1/2	1.5	40	8	20
8	50,000	16,666	2	1.5	40	8	20
9	60,000	20,000	3	1.5	35	8	20
10	75,000	25,000	3	1.5	35	8	20
10a	90,000	30,000	3	1.5	35	8	20
11	100,000	33,333	5	1.2	35	8	20
12	120,000	40,000	5	1.2	35	8	20
13	150,000	50,000	5	1.2	35	8	20
14	180,000	60,000	5	1.2	35	8	20

NOTE: All values shall be not less than those shown except where a range is shown, the values shall be not more than the minimum and not less than the maximum figures shown.

3.4.2 Roll-assemblies. Roll assemblies shall consist of one or more pedestals, axles, and bearings. These items shall be of such materials, size, quantity, alignment, and load carrying capacity as required to meet all performance requirements of the turning rolls, including the loading, turning,

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and unloading of the workpiece. Unless otherwise specified, the rolls shall have rubber and/or urethane tired surfaces. When specified (see 6.2), rolls shall have steel tired surfaces. Rubber or urethane tired rolls shall be provided with a means for tire replacement. When specified (see 6.2), rubber and/or urethane tired rolls shall be provided with steel overload protection rings.

3.4.3 Roll drive mechanism. The type I turning rolls shall be provided with an electronic motor drive mechanism. The mechanism shall provide not less than the turning capacity specified in table I. Drive mechanisms shall be enclosed where feasible, and shall provide proper lubrication for continuous performance. The drive system shall provide a variable surface speed to the driven rolls, with a range as specified in table I. The drive system shall hold the workpiece in a stationary position in the event of power failure.

3.4.3.1 Tilting type turning rolls. The type III tilting turning rolls shall be provided with a tilting base and means to allow for rotating the workpiece 360 degrees. The type III turning rolls shall be capable of tilting the workpiece not less than 45 degrees for longitudinal and end welding. The type III turning rolls shall be provided with surface speed of not less than that specified in table II. The turning rolls and shall be powered by a variable speed electronic drive through a self-locking wormgear reducer. The rolls shall be reinforced with steel hubs to absorb heavy shock loading and to protect the gears from overload.

TABLE II. Size and capacities.

Turning roll type III size	Turning capacity (pounds) type III only	Motor horsepower (HP) type III only (minimum)	Turning roll wheel surface speed inches per minute (IPM)		Workpiece diameter (inches)	
			minimum	maximum	minimum	maximum
1	2,000	1/8	1.0	44	2	48
2	3,000	1/2	1.5	48	6	48
3	5,000	1/2	1.5	60	3	72
4	10,000	1	1.5	60	15	72
5	20,000	3/4	2.0	50	15	72
6	30,000	1-1/2	2.0	50	4	13

NOTE: All values shall be not less than those shown except where a range is shown, the values shall be not more than the minimum and not less than the maximum figures shown.

3.4.4 Car. The type III turning rolls shall be provided with a car. The car for the type III turning rolls shall function as a traversing and support unit. The base of the type III turning roll shall be power driven and shall be capable of supporting the turning rolls and the workpiece. The traverse motion of the type III turning rolls shall be capable of positioning the workpiece for longitudinal welding motion and for longitudinal seam welding along the full length of the workpiece. The traverse motion of the car shall be accomplished by either a silicon controlled rectifier (SCR) solid-state,

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direct current (DC) motor drive or an electronic variable speed drive. The drive shall be provided with a self-locking wormgear reducer. The car shall be provided with not less than four flanged precision machined rolls with self-aligning, permanently lubricated, roller bearings.

3.4.4.1 Tracks. Tracks shall be constructed with a standard 56-1/2 inch gauge rails and shall have means for securing the track to the floor. The track shall be precision machined to commercial standards. The ends shall be milled square to assure smooth fit of additional rails. Additional rails shall be provided as specified (see 6.2). Wipers shall be furnished to remove dirt and other contaminants from the tracks.

3.4.4.2 Traverse capacity. The traverse base of the cars shall allow movement of the complete turning roll, controls, and workpiece along a rail path for longitudinal traverse of the turning roll and the workpiece. Traverse motion shall also be provided for moving the turning rolls on rails between jobs.

3.4.4.3 Car traverse motion. Unless otherwise specified (see 6.2), the traverse motion of the car shall be accomplished with an SCR variable speed electronic drive. Unless otherwise specified (see 6.2), the maximum and minimum rate of traverse motion of the car shall conform to the requirements of table II. Control of the traverse motion shall provide accuracy throughout the traverse range with not more than plus or minus 2 percent deviation from any control set point. The cars shall be equipped with limit switches to prevent overtravel.

3.4.5 Electrical requirements. The turning rolls shall be wired to conform to NFPA 79. Each turning roll shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each turning roll shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-voltage. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The turning roll shall be initially wired as specified (see 6.2).

3.4.5.1 Motors. Motors shall meet the requirements of 3.4.5. Motors shall be rated for continuous duty and have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.5.2 Controls. Controls for operating the powered turning rolls and car traverse motion shall be assembled in a pendant type enclosure. The controls shall be mounted at the end of the turning roll chassis. The controls shall be provided with a multi-conductor control cable not less than 20 feet long. The cable attachment shall be reel type. The control shall include forward, reverse, and stop pushbuttons with "power on" pilot light, rapid traverse, and speed potentiometer. On mechanical drive units, the variable speed control shall mounted on the drive motor unit. When required (see 6.2), the control system shall be furnished with a foot switch, or other remote control devices, as specified. Electronic type drive controls shall be mounted in an enclosure conforming to NEMA ICS I.

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3.4.5.3 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the turning rolls shall meet the requirements of table I for the turning rolls specified. Turning rolls size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the turning rolls shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the turning rolls will fit in its future operating location.

3.6 Performance. The turning rolls shall be capable of rotating a cylindrical workpiece having the total weight and diameter specified in table I. The speed shall be variable, stepless, and reversible. The wheel surface speed shall be accurate to within plus or minus 3 percent of selected speed. Type II turning rolls shall be capable of supporting not less than the weight specified in table I. The type III tilting and turning rolls shall be capable of tilting and turning workpieces as specified in table II and paragraph 3.4.3.1. Each turning roll shall be capable of withstanding the normal usage encountered by this type of equipment during loading, turning, and unloading workpieces.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial turning rolls shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment specified in the following subparagraphs shall be furnished as specified and (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or turning rolls. Optional equipment specified shall be as following

a. Magnetic brakes or electronic motor brakes.

b. Tachometer (accurate to within $\pm$ 2 percent at maximum indication).

c. Rotary type grounding device.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.10 Marking on instruments, control panels, charts, and plates. All words on the turning rolls including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Turning capacity (pounds)
Support capacity (pounds)
Workpiece diameter, minimum and maximum
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Workmanship. Workmanship of the turning rolls and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The turning rolls shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each turning roll prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, and the tests in 4.8. The turning roll shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each turning roll to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the turning roll is installed at its final location. The acceptance test shall consist of the examination in 4.7 And all tests in 4.8. The turning roll shall pass the examination and all tests to be accepted.

4.7 Examination. Each turning roll shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degree to 104 degree Fahrenheit (F) and 20 percent to 95 percent relative humidity.

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4.8.2 Operational test. Each turning roll shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance test. Each turning roll shall be tested for conformance to the performance requirements of 3.6. An actual or simulated support load shall be applied to each turning roll. A turning load shall be applied to each turning roll. The load shall be equal to the maximum load specified in table I for the size unit being tested. The test shall pass all inspections specified in 4.4 and all conformance requirements specified in 4.5 and the examination in 4.7, and be verified during the test period. All performance tests shall be completed without making any adjustments to or performing any maintenance on the turning rolls.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3. shall be as specified and fully described (see 6.2).

4.8.3.2 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each corner of the turning roll. Noise emitted by the turning roll when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type II sound level meter at slow response. The test shall be conducted with the turning rolls operating at intermediate and maximum speeds.

4.8.3.3 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the turning rolls offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided under the provisions of the contract.

4.10 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Turning rolls covered in this specification are intended for use in welding shops, x-ray inspection stations , sandblasting stations, and paint shops for rotating and positioning cylindrical and other workpieces while performing operations in a general downhand style. Turning rolls may be used in automatic and precision welding application.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Capability of handling workpieces with unusual configuration under unbalanced load conditions, if required (see 3.2).

f. System of units for measuring and indicating device (inch pound) (see 3.2.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

i. Noise level requirements, if different (see 3.2.5.2).

j. Environmental protection requirements exceeding EPA/OSHA requirements if different (see 3.2.6).

k. Specify surface finish requirements, if different (see 3.3.3).

l. Painting, if different (see 3.3.5.).

m. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

n. Steel tired surface, if required (see 3.4.2).

o. Overload protection rings, if required (see 3.4.2).

p. Additional rails, if required (see 3.4.4.1).

q. Drive other than SCR variable speed, if required (see 3.4.4.3).

r. Rate of traverse motion, if different (see 3.4.4.3).

s. Voltage requirements, if different (see 3.4.5).

t. Initial voltage, as specified (see 3.4.5).

u. Motors, if different (see 3.4.5.1).

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v. Reel type cable and foot controls or other remote controls, if required, (see 3.4.5.2).

w. Worklight, if required (see 3.4.5.3).

x. Size and capacity, if different (see 3.5).

y. Specify physical size limitation, if required (see 3.5.1).

z. Additional performance features, if required (see 3.6.1).

aa. Specify and fully describe optional equipment, if required (see 3.8).

bb. Specify applicable repair parts, if required (see 3.9).

cc. Lubrication plate, if required (see 3.10.1).

dd. Lubrication plate information, if different (see 3.10.1).

ee. Nameplate, if different (see 3.10.2).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (4.5).

ii. Acceptance test, if different (see 4.6).

jj. Additional performance tests, if required (see 4.8.3.1).

kk. Noise level measurement test, if different (4.8.3.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-T-80131B	MIL-T-80131C		
	Type I	Type II	Type III (new)
	Size 1 (new)	Size 1 (new)	Size 1 (new)
	Size 2a (new)	Size 2a (new)	Size 2 (new)
	Size 7a (new)	Size 7a (new)	Size 3 (new)
	Size 10a (new)	Size 10a (new)	Size 4 (new)
			Size 5 (new)
			Size 6 (new)

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6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

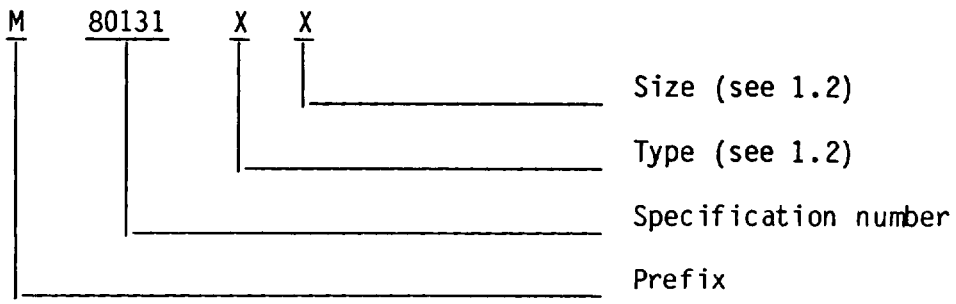
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a turning roll will operate. It does not limit the hazard level of individual turning rolls in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for turning rolls acquired to this specification are created as follows:



6.14 Subject term Key word listing.

Car
Cylindrical workpiece welding
Idler rolls, welding
Rotational workpiece welding
Support rolls welding
Tilting turning rolls, welding
Turning rolls, welding
Tracks
Welding, cylindrical workpiece

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Custodians:
Army - AL
Navy - SH
Air Force -99

Review activities:
Army - AR
Navy - OS
Air Force - 84
DLA - GS

User activities:
Navy - MC, OS

Preparing Activity

DLA - IP

Project 3436-0047

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-80131C	91 10 25

3. DOCUMENT TITLE
TURNING ROLLS, WELDING, POWERED, IDLER, AND TILTING TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)-775-6699	(2) AUTOVON 683-6699
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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